

J DHANA SANTHOSH REDDY

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SUMMARY

AI/ML Engineer specializing in reinforcement learning (RL) environments, large language model (LLM) evaluation, and production agentic systems. Architected an enterprise human-in-the-loop annotation and evaluation platform (Django, React, AWS) serving a ~3,000-person expert network, shipped a production-grade multi-modal (voice + video) AI interviewer with real-time anti-cheating detection, and built polyglot RL environments benchmarking frontier models. Full-stack across Python, PyTorch, Django, React, and AWS, with a research background in robotics perception and motion planning.

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow, LLM Evaluation, LLM-as-a-Judge, Reinforcement Learning (RL) Environments, Reinforcement Learning from Human Feedback (RLHF), Multimodal Machine Learning, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Vision Transformers (ViT), Transformers, Scikit-learn, Hugging Face

LLM & Agent Tooling: Anthropic Claude API, OpenAI API, GPT-4o, function/tool calling, vector embeddings, semantic search, prompt engineering, model-graded rubric scoring, speech-to-text (STT) and text-to-speech (TTS) benchmarking

Evaluation & Data Quality: Inter-rater reliability (Krippendorff's alpha, Cohen's kappa, Gwet's AC1, Intraclass Correlation, Kendall's W), calibration design, annotation QA, dataset forensics, Failure Mode and Effects Analysis (FMEA), A/B and held-out evaluation

Full-Stack & Cloud: Django, Django REST Framework, React, Vite, PostgreSQL, REST APIs, AWS (S3, RDS, Cognito, Elastic Beanstalk, Amplify, WorkSpaces, Secrets Manager, GuardDuty, CloudTrail, WAF, IAM), SOC 2 readiness, CI/CD, Docker, Stripe

Robotics & Perception: ROS/ROS2, OpenCV, CARLA, PointNet++, LiDAR 3D object detection, path planning, motion control, Simulink

Languages & Tools: Python, C++, SQL, JavaScript, MATLAB, Lean, Git, GitHub Actions, Linux, TensorBoard

EXPERIENCE

Verita AI | AI/ML Engineer (Reinforcement Learning & Evaluation) | San Francisco, CA **March 2026 – Present**

- Architected and shipped an internal human-in-the-loop annotation and operations platform (Django/Django REST Framework with ~200 API endpoints, PostgreSQL on AWS RDS, React and Vite single-page applications) that runs labeling and model-evaluation programs for frontier AI clients across a contracted expert network of ~3,000 annotators.
- Built and deployed a production-grade multi-modal (voice & video) AI interviewer with real-time anti-cheating detection: HMAC-authenticated session handoff, an LLM-as-a-judge evaluation pipeline, and composite integrity scoring fused from computer-vision and audio-diarization signals, cutting turnaround time by ~30%.
- Designed and shipped reinforcement learning environments in COBOL, Fortran, and Lean to benchmark frontier models, adding rigorous QA gates and FMEA-driven failure-mode remediation that streamlined evaluation runs and cut debugging time.
- Built an agentic talent sourcing, matching, and outreach engine combining LLM reasoning, structured tool use, and model-graded scoring with semantic search over vector embeddings and rubric-based scoring, ranking high-volume applicant pipelines and surfacing per-criterion rationale.
- Built the platform's quality layer end to end: a dual-review calibration workflow with gap-triggered tiebreak and reviewer averaging, project-scoped quality, throughput, and cost dashboards, bulk task operations with dry-run validation, payout integration, and contributor leaderboards.
- Defined the company's annotation- and evaluation-quality measurement standard using inter-rater reliability statistics (Krippendorff's alpha, intraclass correlation, Gwet's AC1, Kendall's W), and set agreement against human reviewers (Cohen's kappa, Pearson) as a hard launch gate for evaluation products.
- Led cloud security and compliance engineering toward SOC 2 readiness — IAM with enforced multi-factor authentication, AWS GuardDuty, CloudTrail, WAF, Secrets Manager, and Cognito single sign-on — and provisioned 50 isolated virtual desktops (Amazon WorkSpaces with SSO federation) for sensitive client data handling.
- Led development of enterprise-readiness evaluation suites for frontier models, implementing a fine-tuned Vision Transformer (ViT) classifier and a four-classifier ensemble in PyTorch for safety and intellectual-property compliance assessment.
- Developed quality-control and forensic pipelines for large-scale egocentric video dataset procurement — provenance verification, AI-generated content detection, near-duplicate matching, and face detection.
- Developed dual-judge consensus evaluation pipelines (GPT-4o and Claude) alongside multilingual STT/TTS benchmarks, reducing single-model bias and improving evaluation rigor.

- Authored client-facing technical proposals and evaluation-methodology pitches for prospective enterprise partners, translating research capability into scoped, deliverable programs.

Handshake AI, Habitat Inc. (contract) | AI/ML Coding Benchmark Contributor | Remote, USA November 2025 – January 2026

- Engineered benchmark tasks by converting open-source Git commits into structured AI coding challenges with end-to-end Python test suites, enabling consistent evaluation of 30+ models.
- Conducted QA reviews of AI-generated solutions using GitHub pull-request checks and custom lint scripts, documenting failure modes and aligning test coverage with production code-review standards, which reduced post-release defects by 15%.
- Evaluated ML agent trajectories and ideation traces with TensorBoard and custom integrity scripts, detecting misalignment, reward hacking, test-set leakage, and plagiarism (catching 12 violations prior to deployment).
- Assessed ML workflows for experimental rigor using train/dev/test splits, cross-validation, and hyperparameter tuning in Scikit-learn, increasing generalization accuracy by 8%.

Precision Agriculture Lab, University of Maryland | Research Assistant, Deep Learning | College Park, MD May 2024 – February 2025

- Designed and trained Long Short-Term Memory (LSTM) networks to forecast irrigation prescriptions up to 24 hours ahead, achieving RMSE below 0.224 mm for dynamic water management.

PROJECTS

- **Transformer-Based 3D Object Detection:** Engineered an end-to-end transformer pipeline with a novel loss function and PointNet++ embeddings for 3D LiDAR object detection in autonomous vehicles.
- **AI-Powered Learning Playlist Generator:** Built a full-stack application using Tavily search, Google Generative AI, and ElevenLabs text-to-speech to synthesize personalized educational audio content.
- **Fuzzy Adaptive RRT*N Path Planning:** Implemented the FA-RRT*N algorithm in CARLA, reducing computation time by 84% and node exploration by 68% in complex navigation scenarios.

EDUCATION

University of Maryland, College Park | Master of Engineering, Robotics August 2023 – May 2025

Coursework: Multi-Modal Models, AI and Deep Learning, Perception, Path Planning.

SRM Institute of Science and Technology | Bachelor of Technology, Mechatronics Engineering August 2019 – May 2023

PUBLICATIONS

- “Gesture Generation for Robotic Manipulators” — under review, Inderscience Publishers.